

Emily Herbert

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Product and compiler engineer focused on execution systems, AI infrastructure, and developer-facing tooling and interaction design.

Education

2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.

2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.

2025 **Art & Design Intensive**, *Atelier de Sevres*, Paris, France.

Relevant Experience

Oct 2025 – **TAHO (early-stage startup)**.

Apr 2026 Product Engineer - Rust

- Enabled large models (e.g., Qwen3, Stable Diffusion) to run on commodity hardware (e.g., Mac).
- Built distributed AI inference infrastructure using streaming-first model decomposition.
- Developed an IR enabling real-time graph decomposition without full model materialization.
- Contributed to decomposition strategies including critical-node partitioning and planned operator/tensor-level extensions.
- Designed low-latency execution under partial-information constraints.

Aug 2021 - **Fuel Labs (early-stage startup)**.

May 2023 Compiler Engineer - Rust

- Built core components of the Sway smart contract language and compiler (61k+ [GitHub](#) stars).
- Contributed to the type system and type checker (trait resolution, name resolution, generic inference, diagnostics).
- Improved compiler usability through language design (traits, match expressions, generics) and developer feedback.
- Shaped AST design and mid-level compiler architecture across frontend, tooling, and backend teams.

May 2021 – **Google, Madison, WI.**

Aug 2021 Software Engineering Intern - C++

- Implemented load balancing in a high-performance networking library using RMA-based RPC mechanisms.
- Integrated low-level systems for efficient communication with network hardware.

June 2018 - **Northeastern University, University of Massachusetts Amherst, Boston, MA, Amherst, MA.**

May 2021 PhD Researcher - Rust, TypeScript, Python

- Built a serverless function accelerator generating Rust from traced JavaScript executions, without requiring users to adopt Rust.
- Enabled runtime switching from containerized execution to compiled Rust for improved latency and throughput.
- Implemented a full-stack prototype spanning compiler, runtime, networking, and storage layers.

June 2017 – **National Aeronautics and Space Administration (NASA), Langley, VA.**

Aug 2017 Software Engineering Intern - C++

- Built GPS-based geofencing system for autonomous drones across heterogeneous platforms.
- Tested and validated system behavior using simulated hardware environments.

Selected Talks & Publications

2023 Blockchain Language Design panel. *Layer 2 Day at EthDenver*. [[recording](#)]

2022 Developing Smart Contracts in Sway talk. *Layer2Amsterdam*. [[slides](#)]

Sway: A Rust-based Smart Contract Language talk. *EthDenver*. [[slides](#)]

The Rollup Developer Experience panel. *EthDenver*. [[recording](#)]

2021 Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. [[preprint](#), [repo](#)]

2020 Wang Cen, Emily A Herbert, and Peter J Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference*. [[paper](#)]. **Best Contributed Theoretical Paper Finalist.**

Skills

Core Languages, Rust, Sway, TypeScript, Python .

Additional Languages, Haskell, OCaml, Idris, C, C#, C++ .

Focus Areas, Compilers, Programming Languages, Systems Design, Developer Tooling.